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20 March 1987
SUPERSEDING
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18 June 1973
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MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, SMOKE, WP, M156 (METAL PARTS)

This specification is approved for use within Chemical Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the metal parts for the M156 2.75 inch rocket warhead.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using: the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 1340

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Ammunition Lot Numbering.
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

90-1-39 - Warhead, 2.75 Inch Rocket, Smoke, WP, M156 (Metal Parts).

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

8837835 - Pallet for Projectile, Metal Parts.
8861520 - Marking Diagram for Pallets for Projectile, 75MM to 120MM.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components. All materials and components cited on Drawing 90-1-39 and subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.2 Assembly. All components shall be assembled in accordance with Drawing 90-1-39.

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3.3 Leakage and distortion.

3.3.1 Warhead (hydrostatic). The warhead (Drawing 90-1-39) shall not leak or show evidence of distortion when subjected to an internal pressure of 5,000 ± 50 pounds per square inch gage (psig) for a period of 5 to 7 seconds at ambient temperature when tested as specified in 4.4.4.1.

3.3.2 Warhead (air). The warhead accepted for hydrostatic leakage shall not leak air when subjected to an internal air pressure of 100 ± 10 psig for a period of 15 to 20 seconds at ambient temperature when tested as specified in 4.4.4.2.

3.3.3 Burster tube casing. The burster tube casing shall not leak air when subjected to an internal pressure of 100 ± 10 psig for a period of 15 to 20 seconds at ambient temperature when tested as specified in 4.4.4.3.

3.3.4 Brazed joint. When tested as specified in 4.4.4.4, the sectioned and peeled brazed joints shall be free of entrapped slag, unwetted areas, or unbonded areas, for a minimum of 50 percent of the aggregate area of the faying surfaces. Half the area of the adapter joint may be excluded from examination. There shall be no leakage paths through the brazed joints. This will be assured by the presence of a bonded area not less than 1/8 inch wide that extends 360° around each joint.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of warheads and burster tube casings shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The components shall be free from foreign matter (oil, grease or dirt) and damage such as dents, cracks or deformation.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not

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relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 15 warheads and 15 burster tube castings shall be manufactured using the same methods, materials, equipment, and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. The preproduction samples shall be examined and tested in accordance with the classification of defects (4.4.3.3). The AQLs do not apply.

4.3.2.2 For destructive tests. Each brazed joint in the preproduction sample of warheads shall comply with 3.3.4 when tested as specified in 4.4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction samples shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The acceptance number is zero. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168, and shall contain warhead components of one interfix number, produced by one manufacturer, in one unchanged process, at one plant, from the same materials in accordance with the same drawings, the same drawing revision, the same specification and the same specification revision.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive test. During the production of the first lot, one body assembly (Drawing 90-1-39) shall be randomly taken from each shift's production. After the first lot, six body assemblies shall be randomly taken from each succeeding lot.

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. The sample components shall be examined and tested in accordance with the classification of defects (4.4.3.3) and MIL-STD-105.

4.4.3.2 For destructive test. Prior to the application of protective coating and after air test (4.4.4.2), the brazed joints of the sample from the first lot shall be tested and examined in accordance with 4.4.4.4 for compliance with 3.3.4. In the event that a defective brazed joint is found, the contractor shall be required to provide written evidence to the Government of corrective action taken to preclude a recurrence of this defect prior to proceeding with production, and shall make a lot in accordance with the corrective action. Production shall not be initiated until test samples meet the requirements of 3.3.4.

4.4.3.3 Classification of defects.

(a) Adapter (Drawing 90-1-40).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Dimension 1.125 incorrect
102		Overall length incorrect
Minor:	AQL 2.5 percent defective	
201		Dimension 0.130 incorrect
202		Dimension 0.750 incorrect
203		Dimension 0.240 incorrect
204		Workmanship (see 3.5)

(b) Base (Drawing 90-1-41).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Dimension 1.92 incorrect
102		Dimension 2.23 incorrect

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(c) Casing, burster tube (Drawing 90-1-28).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100 percent inspection	
101	Leakage	4.4.4.3
102	Inside diameter incorrect	
Major:	AQL 0.65 percent defective	
103	0.803 diameter and radius not continuous	
104	Taper incorrect	
105	0.400 dimension incorrect	
Major:	AQL 1.0 percent defective	
106	Surface finish incorrect (press fit area)	
107	7.475 dimension incorrect	
108	0.750 diameter incorrect	
109	0.652 dimension incorrect	
110	0.090 dimension (bottom thickness) incorrect	
111	0.8035 diameter incorrect	
112	Check concentricity of 0.8035 diameter to datum A	
113	Straightness of tube, inside contour and depth (note 4 on drawing)	
Minor:	AQL 2.5 percent defective	
201	0.06 radius incorrect	
202	0.75 dimension incorrect	
203	Workmanship (see 3.5)	
204	Wall thickness incorrect	

(d) Warhead, 2.75 inch rocket, smoke, WP, M156, (metal parts) (Drawing 90-1-39).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	100 percent inspection	
101	Hydrostatic leakage	4.4.4.1
102	Air pressure leakage	4.4.4.2

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Categories		Defects
Major:	AQL 0.40 percent defective	
103		Surface finish of 0.7985 diameter and 0.090 lead in and blend radius incorrect, surface analyzer with motortrace
104		Concentricity of 0.7985 diameter of surface D incorrect
Major:	AQL 0.65 percent defective	
105		Diameter 0.7985 incorrect
106		Diameter 2.225 incorrect
Major:	AQL 1.0 percent defective	
107		Thread 1-7/16-16 UN-2B incorrect
108		2.505 dimension incorrect
109		10.395 dimension incorrect
110		Concentricity 1.38 from nose incorrect
111		Diameter 2.77 (surface A) incorrect
112		Pitch or minimum major diameter of rear thread incorrect
113		Concentricity of 2.225 diameter to 2.77 diameter incorrect
114		Diameter of bourrelet after painting
115		Concentricity of bourrelet diameter with outside diameter forward of bourrelet
116		Concentricity of 2.225 diameter to surface F incorrect
117		Point of tangency of 0.7985 diameter and the radius not concentric and without discontinuity - reverse taper (press fit surface)
118		Perpendicularity of body to surface A incorrect
119		Profile incorrect
120		Combination tolerance of concentricity to datum A and perpendicularity to surface C measured 0.25 and 2 inches from nose incorrect
121		Workmanship (see 3.5)
Minor:	AQL 2.5 percent defective	
201		Diameter 2.275 incorrect
202		Concentricity of diameter 1.455 to surface D incorrect
203		Protective finish damaged
204		Pitch diameter of rear thread maximum incorrect

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(e) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Strapping missing, loose or broken
102	Pallet damaged
103	Marking incorrect or illegible

4.4.4 Tests.

4.4.4.1 Warhead (hydrostatic). The warhead shall be subjected to the specified pressure (using soluble oil and water mixture) for the required period of time. Accepted warheads shall be stamped with the letter "H", and the components shall be thoroughly cleaned and dried prior to the air test.

4.4.4.2 Warhead (air). After acceptance for hydrostatic leakage, and cleaned and dried per 4.4.4.1, the warhead shall be subjected to the specified pressure and completely submerged in water for the required time. Any indication of leakage shall be cause for rejection. Acceptable assembled components shall be stamped with the letter "A".

4.4.4.3 Burster tube casing. The burster tube casing shall be subjected to the specified pressure and completely submerged in water for the required period of time. Any indication of leakage shall be cause for rejection.

4.4.4.4 Brazed joint. The brazed joints of the sample warheads shall be sectioned and peeled from the warhead. The brazing and the solder groove on the warhead shall then be examined for compliance with 3.3.4.

5. PACKAGING

5.1 Cleaning and preservation. Warheads shall be clean and free of corrosion. Critical surfaces shall be preserved in accordance with applicable drawings and MIL-P-116.

5.2 Body packing, level C. The warheads shall be packed in accordance with Drawing 8837835.

5.3 Casing, burster tube, level C. The burster tube casing shall be packaged, packed, and marked in accordance with MIL-STD-1169.

5.4 Marking. Shipping containers shall be marked in accordance with Drawing 8861520.

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6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) The shipment group size and marking (when required).
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Subject term (key word) listing.

Explosive warhead
Rocket
Rocket warhead
Smoke
Warhead
WP

Custodians:

Army - EA
Air Force - 99

Review activity:

Air Force - 84

Preparing activity:

Army - EA

Project No. 1340-0703

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-60307C	WARHEAD, 2.75 INCH ROCKET, SMOKE, WP, M156 (METAL PARTS)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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